# Gradient Descent and Optimization

## Section 1: Loss Function Derivatives and Stationary Points

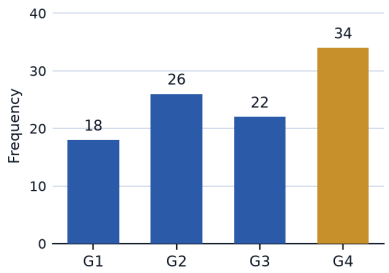
**1** Find the stationary point (global minimum) of the scalar mean squared error loss $L(w) = 3(w-4)^2 + 7$.

**a.** $w = 4$

**b.** $w = 7$

**c.** $w = 3$

**d.** $w = 0$




**2** For the loss function $L(w) = w^4 - 4w^2 + 6$, find all values of $w$ where the gradient is zero.

**a.** $w = 0,\ \pm\sqrt{2}$

**b.** $w = \pm 2$

**c.** $w = 0,\ \pm 2$

**d.** $w = \pm 1$

**3** In optimization, a critical point has $L'(w) = 0$ and $L''(w) = -8$. What type of point is this?

**a.** Local Maximum

**b.** Local Minimum

**c.** Saddle Point

**d.** Inflection Point

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## Section 2: Multivariate Gradients and Directional Derivatives

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**4** Compute the gradient vector $\nabla L$ of $L(w_1, w_2) = 3w_1^2 + 5w_2^2 - 2w_1w_2$ evaluated at $(w_1, w_2) = (2, 1)$.

**a.** $[10, 6]^T$

**b.** $[12, 10]^T$

**c.** $[8, 8]^T$

**d.** $[14, 4]^T$

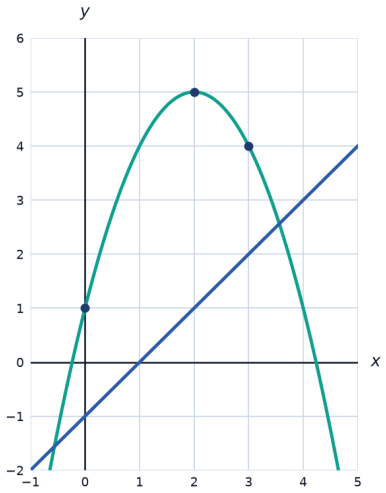
**5** In which direction does the negative gradient vector $-\nabla L(\mathbf{w})$ point?

**a.** The direction of steepest decrease of the loss function

**b.** The direction of steepest increase of the loss function

**c.** Orthogonal to the level curves in the direction of highest loss

**d.** Along the tangent line of the contour




**6** Compute the directional derivative of $f(x, y) = x^2y$ at $(1, 2)$ in the unit direction $\mathbf{u} = [0.8, 0.6]^T$.

**a.** 3.8

**b.** 4.2

**c.** 2.6

**d.** 5.0

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## Section 3: Gradient Descent Step Updates

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**7** A model weight has value $w_0 = 5.0$. If $\frac{\partial L}{\partial w} = 4.0$ and learning rate is $\eta = 0.1$, find the updated weight $w_1$.

**a.** 4.6

**b.** 5.4

**c.** 4.0

**d.** 4.9

**8** Consider the quadratic loss $L(w) = \frac{1}{2}aw^2$ with $a = 10$. What is the maximum learning rate $\eta$ before gradient descent diverges?

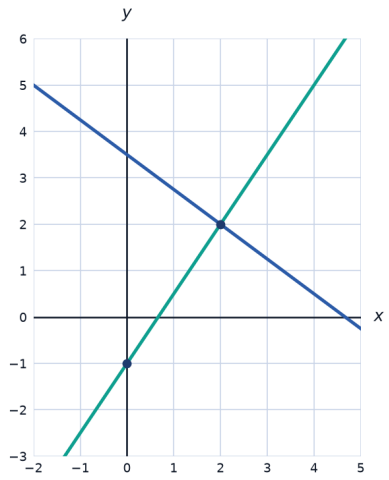
**a.** $\eta < 0.2$

**b.** $\eta < 0.1$

**c.** $\eta < 0.05$

**d.** $\eta < 1.0$

**9** Perform one vector gradient descent update from $\mathbf{w}_0 = [1, 2]^T$ with $\nabla L = [2, -4]^T$ and $\eta = 0.25$.




**a.** $[0.5, 3.0]^T$

**b.** $[1.5, 1.0]^T$

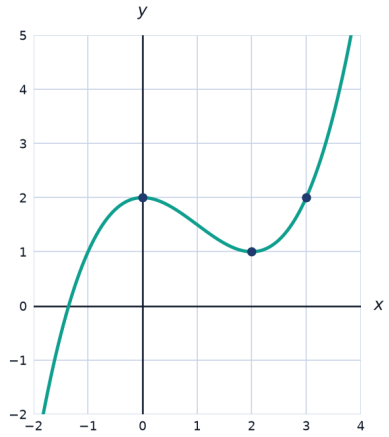
**c.** $[0.5, 1.0]^T$

**d.** $[1.0, 3.0]^T$

## Section 4: Momentum and Advanced Optimizers

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**10** In Polyak momentum, the velocity update is $\mathbf{v}_{t+1} = \beta \mathbf{v}_t + \eta \nabla L$. If $\beta = 0.9$, $\mathbf{v}_0 = 2.0$, $\eta = 0.1$, and $\nabla L = 4.0$, find $\mathbf{v}_1$.




**a.** 2.2

**b.** 2.0

**c.** 1.8

**d.** 2.6

**11** Why does the Adam optimizer maintain running exponential averages of both first moments (mean) and second raw moments (uncentered variance)?

**a.** To adapt per-parameter learning rates inversely proportional to gradient magnitude

**b.** To guarantee convergence to the global minimum in non-convex losses

**c.** To replace backpropagation with finite differences

**d.** To eliminate the need for mini-batch sampling

**12** What is the condition on the Hessian matrix $\mathbf{H} = \nabla^2 L(\mathbf{w})$ for a loss function to be strictly convex everywhere?

**a.** $\mathbf{H}$ is positive definite everywhere (all eigenvalues $> 0$)

**b.** $\mathbf{H}$ has determinant equal to zero

**c.** $\mathbf{H}$ has trace equal to 1

**d.** $\mathbf{H}$ is skew-symmetric

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